O-C plug-in

Overview

This plug-in computes **O-C** (observed minus computed) values for times of light-curve extrema and displays an O-C diagram. A fixed test ephemeris (period and epoch) is compared against observed times of maximum (or minimum) light to reveal timing errors.

Primary reference: Grant Foster, *Analyzing Light Curves*, chapter 13 (AAVSO Variable Star Analytics; clock analogy in Tables 13.1–13.2).

Two plug-ins are provided.

Plug-in	Menu	Role
O-C	**Tools → O-C…**	Main O-C tool (includes **Export CSV…** in results)
O-C demo data	**File → O-C demo data…**	Optional Foster clock light curves

The O-C tool can be opened from the **Tools** menu, even when no observations have been loaded, since other than a **From observations** option, an **Imported timings file** option allows a previously exported/created timings file to be opened instead.

Published help (PDF): [O-C on aavso.github.io].

Installation

1. In VStar, choose **Tools → Plug-in Manager…**
2. Select and install **O-C** (required). Optionally install **O-C demo data** for Foster tutorial light curves.
3. **Close and restart VStar** so new menu items appear.

See also [AAVSO VStar Plug-in Library].

Optional: **File → Preferences… → Plug-in Settings** to check the plug-in download URL matches your VStar version (see the [VStar wiki installation recipes]).

Quick start

1. **Tools → O-C…**

2. Set **data source**, ephemeris (period and epoch), and other options in the parameter dialog.
3. Click **OK** (a file chooser opens if you chose **Imported timings file**).
4. Review the results dialog: **O-C diagram**, **Data table**, and **Fit summary**.

See **Examples** for walkthroughs. **Example 2** uses the optional demo observation source; **Appendix A** covers Foster validation (including imported timings for all six clocks).

Examples

These walkthroughs build on each other:

Example	Purpose
1	Real AID light curve → measure timings → export CSV → re-import timings (typical workflow on your own data)
2	Bundled Foster clock 2 demo → validate the tool against a known O-C offset (no download or real star required)

Use **Appendix A** to validate the other Foster clocks or to import pre-computed timing files.

Example 1 — O-C from loaded observations (AID data)

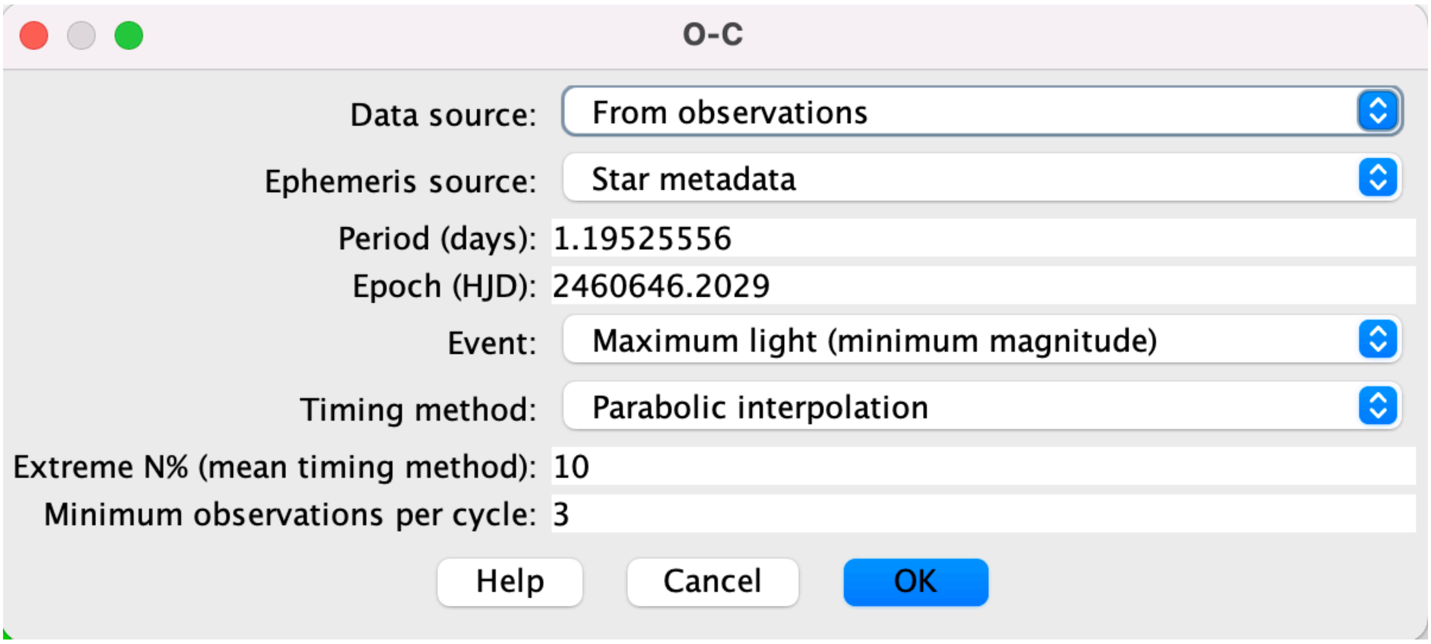
Use this when you have a light curve in VStar and want times of eclipse minima (or maxima) measured from the data.

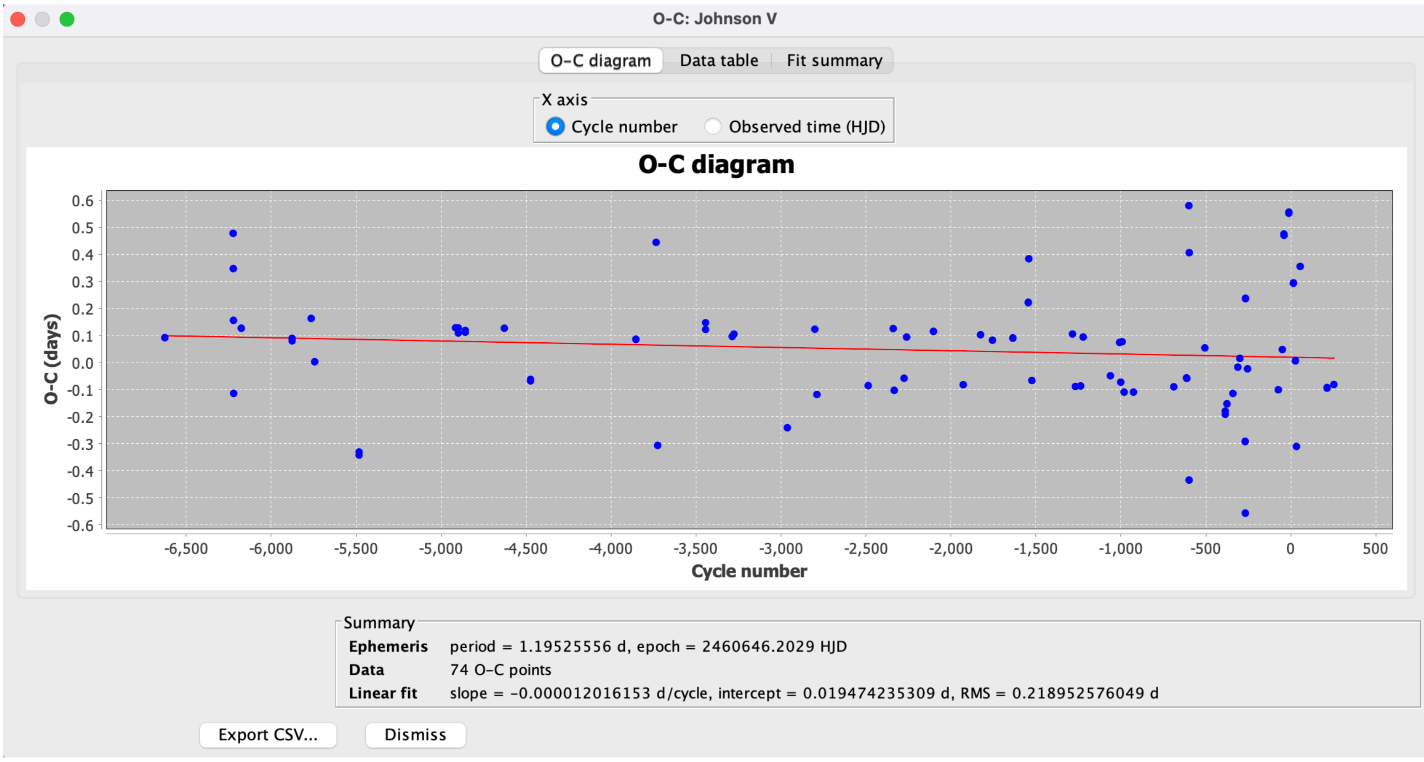
Suggested target: RZ Cas (AUID `000-BBF-490`), a bright Algol-type eclipsing binary (EA). Period ≈ **1.20 d** (VSX), good AAVSO **V** coverage, and a deep primary minimum suited to parabolic timing.

Suggested JD range: 2452700 – 2461000 (about 22 years; calendar **2003-03-01** through **2025-11-20**). The AID holds **9000+ V** observations in this span. In the load dialog, enter those minimum and maximum JD values and select **V** only (clear other series checkboxes).

1. Load the star from the AAVSO International Database (**File → New Star from AAVSO Database…**): star name **RZ Cas**, JD range and **V** series as above.
2. Optional: refine period and epoch with **period analysis** or a **phase plot** if you plan to use **Phase plot** as the ephemeris source (see **Appendix B**). For this example, **Star metadata** (VSX period and epoch) is enough to start.
3. **Tools → O-C…**
4. **Data source:** From observations.
5. **Ephemeris source: Star metadata** (VSX values loaded with the star).
6. **Event:** Minimum (primary eclipse). **Timing method:** Parabolic interpolation. **Minimum observations per cycle:** 3 (defaults are fine if already set).

7. Click **OK**, select the **V** series if prompted.
8. On the **O-C diagram**, check the trend. Open **Fit summary** for linear (and, if applicable, quadratic) interpretation text. Over many years, EB O-C can show real long-term effects (apsidal motion, mass transfer); see **Further reading** below.
9. Click **Export CSV…** in the results dialog and save the file (e.g. `rz_cas_timings.csv`). The export lists each timed eclipse (`Cycle`, `O_time`, optional `OC_sigma`) plus O-C columns and fit metadata in `#` comment lines.
10. **Tools** → **O-C…** again (no star loaded is fine).
11. **Data source:** Imported timings file → **OK** → select the CSV from step 9.
12. Leave **Period**, **Epoch**, and **Event** at their defaults — an O-C export CSV supplies these from its `# period=…, epoch=…` comment and `Event` column. Click **OK** again. The O-C diagram should match step 8 without re-processing the AID light curve.



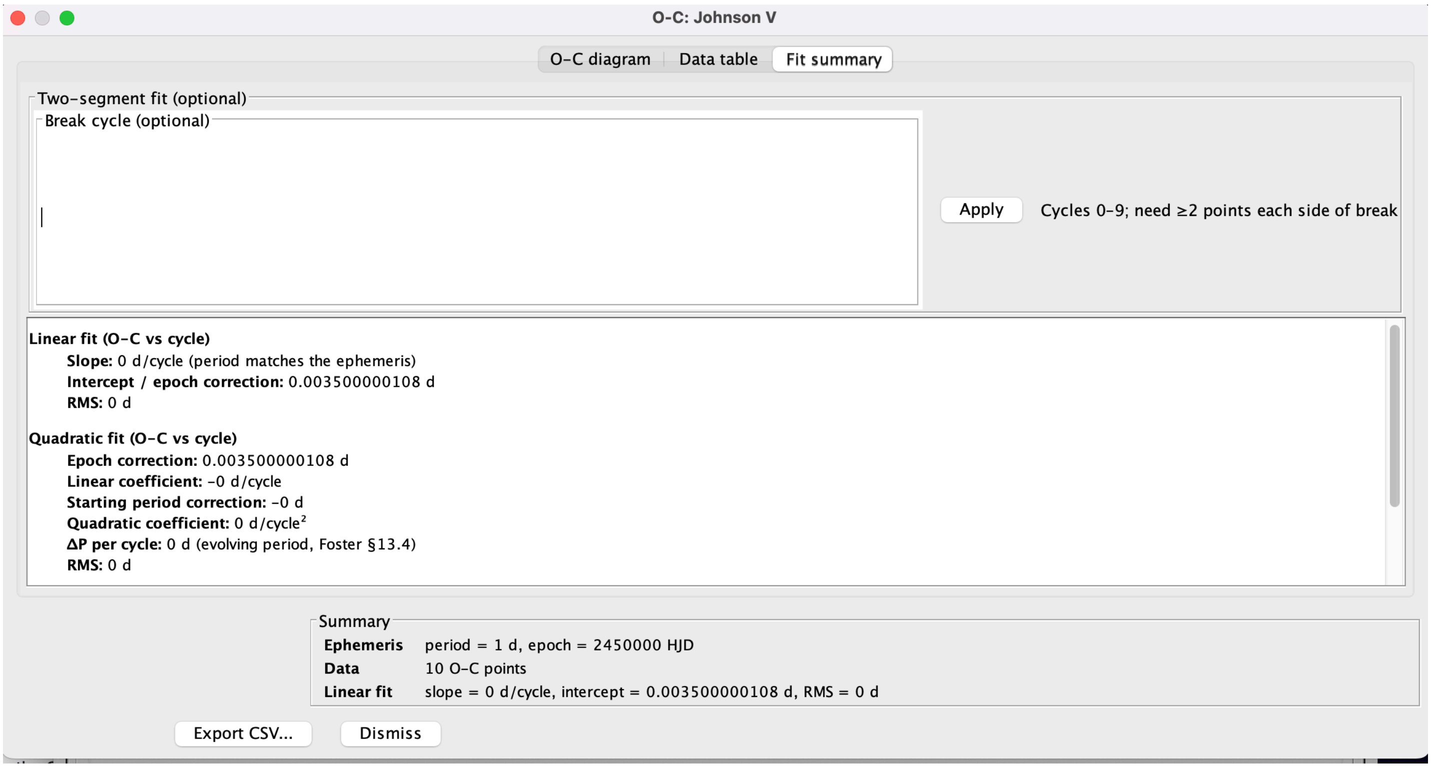


Example 2 — O-C from demo light curve (Foster clock 2)

Use this to walk through Foster's clock analogy with bundled synthetic data — no real star or downloaded timing file required. This example loads Foster clock 2 light curves whose maxima match Table 13.1 and should yield a flat O-C offset of **+0.0035 d** (Table 13.2).

1. Install the optional **O-C demo data** plug-in if needed (see **Installation**), then restart VStar.
2. **File → O-C demo data…**
3. **Demo scenario: Foster clock 2 — epoch offset → OK**.
4. Read the load message. Note the **suggested ephemeris** for Foster test theory: **P = 1.0 d**, **epoch = 2450000.0**, and the expected flat O-C ≈ **+0.0035 d**.
5. **Tools → O-C…**
6. **Data source:** From observations.
7. **Ephemeris source: Star metadata** (pre-fills the suggested period and epoch from step 4). **Event:** Maximum. **Timing method:** Parabolic interpolation (defaults are fine).
8. Click **OK**, select the **V** series if prompted.
9. Review results: O-C should be flat near **+0.0035 d** on every cycle (Foster Table 13.2, clock 2). Open **Fit summary** for the linear-fit interpretation.

To validate the other Foster clocks from light curves, repeat steps 2–9 with a different **Demo scenario** (see **Appendix A**, demo scenario mapping). To validate from pre-computed times instead, use **Appendix A**, Method A.



Further reading — X Tri, eclipsing binaries, and other real stars

Foster discusses real variables (e.g. X Tri, SU Vir, Z Aur) with O-C diagrams built from published **times of maxima** over many cycles. Those examples need literature ephemerides and timing tables; they are not bundled with this plug-in. **Example 2** walks through Foster clock 2 via the demo source; use **Appendix A** for all six clocks (demo light curves or imported timings). For real stars, obtain timings from the literature or measure extrema from your own observations, then run O-C with **From observations** or **Imported timings**.

For **eclipsing binaries**, decades of **eclipse timings** can reveal **apsidal motion** (sinusoidal O-C from general relativity and tidal distortion) or **period change** from mass transfer; those effects need much longer baselines than Example 1.

Results dialog

After O-C is computed, a modeless results window opens with three tabs:

Tab	Contents
O-C diagram	O-C points (optional error bars); toggle **Cycle number** or **Observed time** on the x-axis
Data table	Cycle, observed time, computed time, O-C, uncertainty
Fit summary	Interpretation text for fits (see below)

Linear fit — computed automatically when there are at least two O-C points; shown as a line on the diagram and described on **Fit summary**.

Quadratic fit — there is **no** quadratic option in the parameter dialog. When there are at least three O-C points, a quadratic least-squares fit of O-C versus cycle is computed automatically and described **only on Fit summary** (text, not drawn on the chart). Use this for curved O-C trends (Foster clock 6 / evolving period).

Two-segment fit — optional. On **Fit summary**, enter a **break cycle** and click **Apply** (see **Appendix B**). Needs at least four O-C points and at least two on each side of the break.

Export CSV… on the results dialog saves O-C points and fit metadata to a file.

Components (summary)

O-C tool

Parameter dialog uses a compact two-column layout. **Ephemeris source** is a drop-down only (not typable); edit **Period** and **Epoch** in their fields.

After **OK**, choose an observation series or a timings file depending on **Data source**; the results dialog opens next. Use **Export CSV…** there to save results.

O-C demo data (observation source)

Optional. Loads synthetic light curves whose **maximum timings** match Foster Table 13.1 (clocks 1–6). Bump shape is identical for every clock; only timing differs. Then run **Tools → O-C…** with **From observations** and the suggested ephemeris from the load message (P = 1 d).

Appendix A — Validation against Foster (Tables 13.1–13.2)

Foster's six test clocks use theory **P = 1 day**, **epoch = cycle 0**. Computed time C*n = n. O-C = O*n – C_n

(Table 13.2).

Reference O-C values (Table 13.2)

Cycle	Clock 1	Clock 2	Clock 3	Clock 4	Clock 5	Clock 6
0	0	+0.0035	0	–0.0014	0	0
1	0	+0.0035	+0.0021	–0.0014	–0.0014	0
2	0	+0.0035	+0.0042	–0.0014	–0.0028	+0.0007
3	0	+0.0035	+0.0062	–0.0014	–0.0042	+0.0021
4	0	+0.0035	+0.0083	**+0.0257**	–0.0056	+0.0042
5	0	+0.0035	+0.0104	+0.0257	–0.0069	+0.0069
6	0	+0.0035	+0.0125	+0.0257	–0.0062	+0.0104
7	0	+0.0035	+0.0146	+0.0257	–0.0056	+0.0146
8	0	+0.0035	+0.0167	+0.0257	–0.0049	+0.0194
9	0	+0.0035	+0.0188	+0.0257	–0.0042	+0.0250

Demo scenario mapping

Foster clock	Demo menu label	Expected in plug-in
1	Foster clock 1 — correct ephemeris	Flat O-C ≈ 0
2	Foster clock 2 — epoch offset	Flat O-C ≈ +0.0035 d
3	Foster clock 3 — period error	Slope ≈ +0.0021 d/cycle
4	Foster clock 4 — epoch jump	Step at cycle 4; break **4**
5	Foster clock 5 — period change	Slope change; break **6**
6	Foster clock 6 — slowing clock	Curved O-C; quadratic on Fit summary

Method A — imported timings

For each clock, download the timing file (links below), then:

1. **Tools → O-C… → Imported timings file**.
2. **Period:** 1.0 d, **Epoch:** 2450000.0.

3. Compare O-C and **Fit summary** to Table 13.2.

Method B — demo observation source

See **Example 2** for Foster clock 2. For any clock:

1. **File** → **O-C demo data…** → pick a Foster clock (labels match the demo scenario mapping table above).
2. **Tools** → **O-C…** → **From observations**; use the suggested ephemeris from the load message (**Star metadata** pre-fills period and epoch).
3. Compare O-C and **Fit summary** to Table 13.2.

Foster timing files (Table 13.1 → times)

Epoch **2450000.0**, period **1.0** d for O-C tool entry. Download from aavso.github.io (published with plug-in docs; same files in `plugin/doc/foster/` in the repo).

Clock	Scenario	Download
1	Correct ephemeris	foster_clock1.txt
2	Epoch offset	foster_clock2.txt
3	Period error	foster_clock3.txt
4	Epoch jump (try break cycle 4)	foster_clock4.txt
5	Period change (try break cycle 6)	foster_clock5.txt
6	Slowing clock (quadratic on Fit summary)	foster_clock6.txt

Appendix B — Reference

Ephemeris source

When **From observations** is selected:

UI label	Meaning
Phase plot	Period and epoch from the **active phase plot** (View → Phase Plot, or Period Analysis → New Phase Plot)
Star metadata	Period and epoch from the **loaded star record** (often from an AID download or other source that supplies catalogue fields)
Manual entry	Type period and epoch yourself

Disabled for **Imported timings** — enter period and epoch directly, unless the file is an O-C export CSV (metadata is read from the file).

Imported timings file format

One timing per line: `cycle time [sigma]` or `time [sigma]`. Whitespace, comma, or semicolon separators. `#` comments allowed.

Use **one consistent time system** for observations, epoch, and imported timings (JD, HJD, BJD_TDB, etc.). The plug-in does not convert between systems; mix them and O-C will be wrong. Homogenise with VStar's HJD/BJD tools first if needed.

You can also re-open a file saved with **Export CSV…** from a previous O-C run. The tool reads timings from the data rows and, for export files, fills in **Period**, **Epoch**, and **Event** from the `# period=…, epoch=…` comment and the `Event` column (leave those fields blank in the parameter dialog). Export columns are `O_time` and `C_time` (not a specific JD flavour).

Timing methods (From observations)

- **Parabolic interpolation** — parabolic refinement at discrete extremum.
- **Mean JD of extreme N%** — mean time of brightest N% in each cycle.
- **From current model function** — analytic extrema from a JD-based model.

Event type

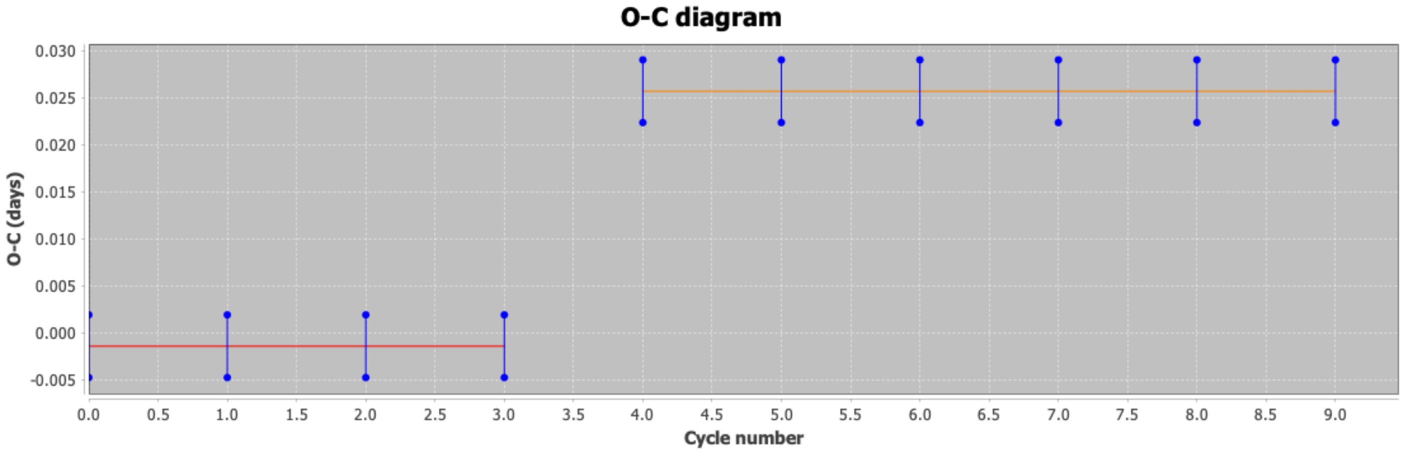
- **Maximum** or **Minimum**
- **Both maxima and minima** — for eclipsing binaries on real data (not used in Foster's six-clock tutorial)

Interpreting O-C diagrams

Pattern	Likely cause
Flat at 0	Ephemeris matches
Flat, offset	Epoch wrong, period OK
Linear slope	Period wrong
Broken line, parallel segments	Epoch jump
Broken line, different slopes	Period change
Curved (parabolic)	Evolving period → Fit summary quadratic

Two-segment fit

On **Fit summary** after results: enter **break cycle**, click **Apply**. Needs ≥4 O-C points and ≥2 per segment.



Parameters (initial dialog)

Parameter	Description
Data source	Observations or imported timings
Ephemeris source	Pre-fill (observations only)
Period, epoch	Test ephemeris (required; same time system as the data)
Event	Maximum, minimum, or both
Timing method, extreme N%, min obs	Observations only

Fit summary (after results)

Output	When
Linear fit	≥2 O-C points; drawn on chart
Quadratic fit	≥3 O-C points; **text only** on Fit summary
Two-segment fit	User break cycle + Apply; drawn on chart

References

- Foster, G. *Analyzing Light Curves*, O-C Analysis, ch. 13, pp. 263–282
- AAVSO Variable Star Astronomy, ch. 13: [Variable Stars and O–C Diagrams]
- [VStar plug-in library]
- [VStar issue #93]

Version history

- **1.8** — Neutral time labels (`O_time` / `C_time` , Epoch, Observed time); docs stress a consistent time system (JD/HJD/BJD/…) rather than assuming HJD. Removed unused O-C CSV observation-sink plug-in (export is results-dialog only).
- **1.7** — Added screenshots for parameter dialog, results diagram, Fit summary, and two-segment fit.
- **1.6** — User-facing rename: **O-C** (Tools menu) and **O-C demo data** (File menu); plug-in group **Timing**.
- **1.5** — Example 2 uses O-C demo data (Foster clock 2); Foster `.txt` downloads remain in Appendix A Method A only.
- **1.4** — User-facing doc restructure: Plug-in Manager install, examples, appendices, Foster timings on aavso.github.io (absolute URLs), screenshot placeholders.
- **1.3** — Foster Table 13.1 demo timings; six clocks only; compact dialog; general tool; break cycle on Fit summary.
- **1.2** — Imported timings, CSV export, quadratic fit, BOTH event type, demo observation source.
- **1.1** — Model timing, linear/two-segment fits, error bars.
- **1.0** — Initial observation-based O-C plot and table.